



Course Information Form

This Course Information Form provides the definitive record of the designated course

Section A: General Course Information

Course Title	BSc (Hons) Software Engineering/ Software Engineering (with Professional Practice Year)/ Software Engineering (with Foundation Year) - COPY
Final Award	BSc (Hons)
Route Code	BSCESAAB/BSSEFAAF/BSSEPAAF
Intermediate Qualification(s)	
FHEQ Level	6
Location of Delivery	University Square Campus, Luton
Mode(s) and length of study	Full-time over 3 years Full-time with Professional Practice Year over 4 years Part-time pathway typically over 6 years
Standard intake points (months)	October and February
External Reference Points as applicable including Subject Benchmark	QAA Subject Benchmark Statement Computing (March 2022) FHEQ (2014) SEEC Credit Level Descriptors (2021)

Professional, Statutory or Regulatory Body (PSRB) accreditation or endorsement	None
HECoS code(s)	100374
UCAS Course Code	G600

Course Aims	Why study this course The course is broadly based yet enough depth (programming skills, testing skills etc.) so as to give you credible vocational skills in Software Engineering. It covers the entire Software Engineering lifecycle and is taught by staff with both a strong academic research background who are also actively engaged in knowledge transfer (e.g. to local SME's), and who are at the forefront of their field. Educational Aims This course is highly vocational with an emphasis on providing students with practical experience as software engineers. This course starts with fundamentals of software engineering studies, principles of programming, concepts for computational thinking, databases and computer networks in the first year. And then providing a platform you will engage in specialist study that will make you confident with the principles and practices of software engineering. The focus will be on large-scale software design, desktop and mobile development and deployment. You will emphasise the entire life cycle of the system and put the production of quality software systems into management and organisational contexts.
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Course Learning Outcomes	Upon successful completion of your course you should meet the appropriate learning outcomes for your award shown in the table below		
		Outcome	Award
	1	LO1: Understand the role of typical commercial development tools used in software engineering settings and critically appreciate the role of the object oriented paradigm in framing the use of these tools;	BSc (Hons) Software Engineering (all routes)
	2	LO2: Understand and critically evaluate the problems and constraints operating within large scale software engineering projects including the role of standards, methods and tools as well as trust issues;	BSc (Hons) Software Engineering (all routes)
	3	LO3: Critically evaluate state-of -the-art tools and techniques currently used in software engineering;	BSc (Hons) Software Engineering (all routes)
	4	LO4: Critically appreciate the constraints operating within a typical software development lifecycle and identify optimal solutions (reqs. engineering methods, analysis and design tools, implementation, testing, post-release);	BSc (Hons) Software Engineering (all routes)
	5	LO5: Work effectively and appreciate the role of working within a software engineering development team and to be able to explain the conditions necessary for successful team working;	BSc (Hons) Software Engineering (all routes)
	6	LO6: Communicate ideas both in writing and orally according to appropriate academic or professional standards;	BSc (Hons) Software Engineering (all routes)
	7	LO7: Know when and why you need information, where to find it, and how to evaluate, use and communicate it in a concise and ethical manner in specific relation to the needs of particular stakeholders in the software engineering industry;	BSc (Hons) Software Engineering (all routes)
	8	LO8: Research and evaluate information from a number of sources;	BSc (Hons) Software Engineering (all routes)
	9	LO9: Apply formal and informal creativity and critical thinking techniques to the solution of problems.	BSc (Hons) Software Engineering (all routes)
	10	LO10: Demonstrate knowledge and analytical understanding of professional practice by successfully completing an approved period of approved work place practice.	BSc (Hons) Software Engineering (with Professional Practice Year)
Learning and Teaching			

Teaching, learning and assessment strategies

The course structure across levels is implemented as follows:

The first year will cover the fundamentals of the broader subject area (Computer Science) by encompassing the topics of Programming, Modelling, Software Engineering, Databases, Networking and Hardware. In addition the professional development of the students is addressed.

The second year is comprised of four specialist units that deepen the knowledge, understanding and application of the first year. While there is no direct prerequisite, there is a clear logic in structure from the first year to the second year:

Desktop Applications Development and Software Engineering follows the foundations provided in Principles of Programming and Data Structures with a focus on object oriented programming and software development in desktop environment.

Mobile Applications Development extends the experiences of Principles of Programming and Data Structures to mobile platforms.

CIS007-2 Decision Support Systems and Data Mining specialises on software engineering ideas and algorithms.

The final year devotes 60 credits to the project. While the undergraduate project relates to the student working as an individual the 30 credit unit Agile Project Management addresses student interaction within a professional environment. The students have to work in a group and make decisions within professionally arranged project meetings.

In addition, the unit CIS046-3 Software for Enterprise provides a capstone to the subject focussed experience from the second year by critically questioning the role of strategies, paradigms, and deployment of distributed systems, cloud computing and their enterprise applications.

Assessment

You are assessed in a variety of ways. The majority of units are assessed through coursework, group and individual projects, portfolios, essays, presentations or exams. You will also produce software artefacts in the area of your specialism. Constant feedback and advice from a supervisory or unit team will be provided to support you in your work.

At level 4 you are assessed on your understanding of the fundamental concepts of Software Engineering and its application. You are required to comprehend the basic range of intellectual concepts which form the foundations of the subject and application area, and will be assessed on your ability to articulate such concepts in a coherent manner, in a variety of written assessments/written briefs. For example, there will be time constraint programming assignments as well as multiple choice tests.

	At level 5 you are assessed on your ability to apply the basic concepts of the disciplines introduced in level 4 to existing controversies and issues on which there is already a body of research and critical opinion. You also should be able to demonstrate the inter-relationships between critical theory and practice. For example, the units 'Mobile Applications Development' and 'Desktop Applications Development and Software Engineering' will further your understanding of software paradigms with presentable prototypes for future career development. Also new concepts (such as machine learning) are introduced. At level 6 you will be required to demonstrate independent thinking and initiative. This may be in the form of analysing and criticising current approaches and theory within software engineering and programme development. In all cases, you will be expected to show an awareness of the major theories and practices of the discipline. You will progress from well-defined briefs to more open-ended and challenging assessments, which culminate in the project – where you will be given freedom to choose your area of work.
Learning support	The University's comprehensive student support service includes: Student Information Desk, a one-stop shop for any initial enquiries; Student Support team advising and supporting those with physical or learning needs or more general student well being; Study Hub team providing academic skills guidance; Personal Academic Tutoring system; a student managed Peer-Assisted Learning scheme; support from your lecturers
Admissions Criteria	https://www.beds.ac.uk/entryrequirements Approved Variations and Additions to Standard Admission None
Assessment Regulations	https://www.beds.ac.uk/about-us/our-university/academic-information Note: Be aware that our regulations change every year Approved Variations and Additions to Standard Assessment Regulations' n/a

Section B: Course Structure

The Units which make up the course are listed below. Each unit contributes to the achievement of the course learning outcomes either through teaching (T), general development of skills and knowledge (D) or in your assessments (A).

Unit	Unit Name	Level	Credits	Core or Option	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
CIS092-1	Fundamentals of Software Engineering	4	30	Core	T1 T2	T1 T2			T1 T2	T1 T2	T1 T2	T1 T2	T1 T2						
CIS093-1	Mathematics and Concepts for Computational Thinking	4	30	Core	T1 T2	T1 T2		T1 T2	T1 T2		T1 T2								
CIS095-1	Databases and Computer Networks	4	30	Core	T1 T2	T1 T2			T1 T2	T1 T2	T1 T2	T1 T2	T1 T2						
CIS096-1	Principles of Programming and Data Structures	4	30	Core	T1 T2				T1 T2	T1 T2	T1 T2	T2	T1 T2						
CIS007-2	Decision Support Systems and Data Mining	5	30	Core	D1 D2	D1 D2	D1 D2		D1 D2	D1 D2	D1 D2	D1 D2	D1 D2						
CIS051-2	Web Technologies and Platforms	5	30	Core	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2						
CIS099-2	Mobile Applications Development	5	30	Core			D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2						
CIS116-2	Desktop Applications Development and Software Engineering	5	30	Core	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2	D1 D2						
CIS013-3	Research Methodologies and Emerging Technologies	6	30	Core	A1 A2		A1 A2												
CIS017-3	Undergraduate Project	6	30	Core	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2						
CIS046-3	Software in Enterprise	6	30	Core	A1 A2		A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2						
CIS047-3	Agile Project Management	6	30	Core	A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2	A1 A2						

Unit	Unit Name	Level	Credits	Core or Option	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
CIS097-2	Professional Practice Year (Computer Science and Technology)	5	0	Option										A1 A2					

Section C: Assessment Plan

The course is assessed as follows :

BSCESAAB-

Unit Code	Level	Period	Core/Option	Ass 1 Type code	Ass 1 Submit wk	Ass 2 Type code	Ass 2 Submit wk	Ass 3 Type code	Ass 3 Submit wk	Ass 4 Type code	Ass 4 Submit wk
CIS092-1	4	SEM1	Core	WR-I	7	EX-CB	12				
CIS093-1	4	SEM1	Core	IT-PT	6	PJ-ART	13				
CIS095-1	4	SEM2	Core	CW-CS	9	IT-PT	11				
CIS096-1	4	SEM2	Core	IT-PT	8	PJ-ART	13				
CIS099-2	5	SEM1	Core	WR-I	6	CW-EPO	13				
CIS116-2	5	SEM1	Core	PJ-ART	7	EX	13				
CIS007-2	5	SEM2	Core	WR-PR	8	WR-PR	14				
CIS051-2	5	SEM2	Core	CW-CS	8	PJ-ART	13				
CIS013-3	6	SEM1	Core	CW-RW	6	WR-I	12				
CIS046-3	6	SEM1	Core	PJ-ART	12	EX	13				
CIS017-3	6	SEM2	Core	WR-I	8	PR-VIV	12				
CIS047-3	6	SEM2	Core	PR-ORAL	10	PJ-ART	13				

BSSEPAAF- Software Engineering (with Professional Practice Year)

Unit Code	Level	Period	Core/Option	Ass 1 Type code	Ass 1 Submit wk	Ass 2 Type code	Ass 2 Submit wk	Ass 3 Type code	Ass 3 Submit wk	Ass 4 Type code	Ass 4 Submit wk
CIS097-2	5	TY	Option	CW-PO	33						

Glossary of Terms for Assessment Type Codes

CW-CS	Coursework - Case Study
CW-EPO	Coursework - e-Portfolio

CW-PO	Coursework - Portfolio
CW-RW	Coursework - Reflective Writing
EX	Exam (Invigilated)
EX-CB	Computer-based Invigilated Examination
IT-PT	Summative in-class test or phase test
PJ-ART	Coursework - Artefact
PR-VIV	Practical - Viva
WR-I	Coursework - Individual Report
WR-PR	Coursework - Problem Based Report

Administrative Information

Faculty	Creative Arts Technologies and Science
School	School of Computer Science and Technology
Head of School/Department	Paul Sant